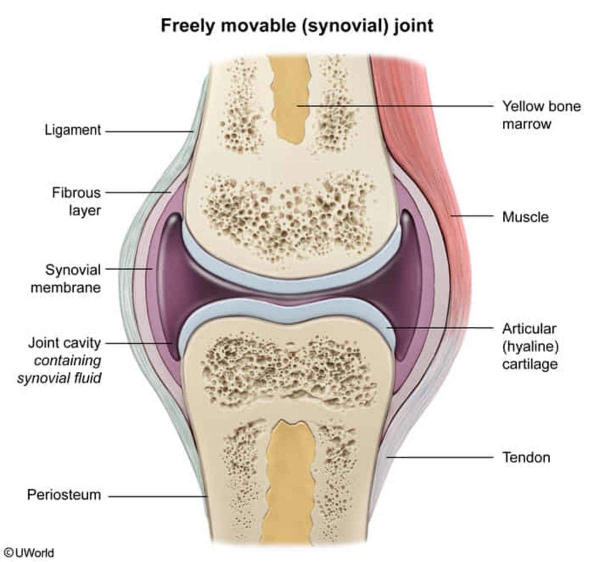
A physician determines that a patient has injured a connective tissue structure connecting two bones within the knee joint. Given this finding, the likely basis of the injury is damage to:

- A. skeletal muscle.
- B. adipose tissue.
- C. a tendon. [15%]
- D. a ligament. [83%]

Correct
82% answered correctly

Explanation:



Joints are specialized components of the vertebrate musculoskeletal system where two (or more) bones articulate (ie, interact). There are several types of joints, including immovable joints (eg, skull sutures) and freely movable joints (eg, knee). Freely movable joints, also known as **synovial joints**, consist of several structures that interact with bone and skeletal muscle:

- Articular (hyaline) cartilage surrounds the ends of bones within the joint, providing a smooth surface that absorbs compressive forces and reduces friction between bones as they interact.
- The joint cavity, located between articulating bones, contains specialized fluid (ie, synovial fluid) that lubricates joint surfaces and protects articular cartilage from excessive friction and damage.
- Fat pads are adipose tissue structures present in some synovial joints (eg, knee and hip joints) that provide additional cushion between bones within the joint.
- **Ligaments** and tendons are rope-like, dense connective tissue structures.

Ligaments attach bones to other bones whereas tendons generally attach *bones to surrounding muscles* **(Choice C)**.

In the given scenario, the patient has injured a connective structure **connecting two bones within the knee**. Because *ligaments* form bone-to-bone connections within joints, the most likely basis of this injury is **damage to a ligament**.

(Choice A) Within a joint, the interaction of skeletal muscles with connective tissue structures facilitates movement. Damage to skeletal muscle may impair mobility within the joint. However, muscle is *not* a connective tissue structure and is *not* directly responsible for bone-to-bone connections.

(Choice B) Adipose tissue is a specialized type of connective tissue that stores fats and cushions surrounding organs. Although adipose tissue comprises the fat pads within some joints, this connective tissue is *not* directly responsible for the attachment of bones to other bones.

Educational objective:
Joints are structures of the musculoskeletal system where bones articulate (ie, interact) and can range in mobility from freely movable to immovable. Within movable joints, strong connective tissue structures called tendons attach muscle to bone whereas ligaments attach bone to bone.

Time spent:0 sec	**QID**:402348
Subject:	**System:**
Biology	Musculoskeletal System